Data Cleaning Assignment

Assignment

PW SKILLS

Dataset Link : Healthcare_Dataset

Q1. Missing Data Identification

Scenario:

The hospital suspects incomplete patient records.

Task:

- Identify missing values in each column
- Calculate percentage of missing data

Q2. Handling Missing Age

Scenario:

Age is critical for medical analysis, but some values are missing.

Task:

- Replace missing Age values with an appropriate method
- Justify your choice (mean/median)

Q3. Handling Missing Treatment Cost

Scenario:

Treatment cost is highly skewed due to expensive treatments.

Task:

- Handle missing Treatment_Cost values
- Choose the correct imputation method and explain why

Q4. Duplicate Patient Records

Scenario:

Some patient records were entered multiple times.

Task:

- Identify duplicate rows
- Remove duplicates
- Compare dataset size before and after

Q5. Invalid Age Values (Data Quality Check)

Scenario:

Some patients have unrealistic age values (e.g., >100 or <0).

Task:

- Detect such records
- Decide whether to remove or correct them

Q6. Outlier Detection (Treatment Cost)

Scenario:

Extreme treatment costs are affecting analysis.

Task:

- Detect outliers using IQR method
- Display number of outliers

Q7. Outlier Treatment

Scenario:

The business team wants to retain all records.

Task:

- Apply capping (Winsorization) on Treatment_Cost
- Use 5th and 95th percentile

Q8. Transformation

Scenario:

Treatment cost is highly skewed.

Task:

- Apply log transformation
- Create a new column
- Compare before vs after distribution

Q9. Time-Based Missing Handling

Scenario:

Admission dates should follow a logical sequence.

Task:

- Sort data by Admission_Date
- Apply forward fill or backward fill where appropriate
- Justify your choice